

PCP Project Problem Statement

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Developing Parallel Fast Fourier Transformation Algorithm

In this project, we will focus on optimizing the Fast Fourier Transformation (FFT) algorithm. The standard Discrete Fourier Transformation (DFT) is computationally intensive with a time complexity of $O(N^2)$. While the standard Cooley-Tukey FFT algorithm significantly improves this to $O(N \log N)$ using a divide-and-conquer approach, we will try to further accelerate it. We will implement and compare two different versions of the FFT algorithm.

We outline the following possible approaches for implementing and comparing FFT algorithms, which may be refined or change as the project evolves:

- *Idea 1:*
 - Algorithm 1: A standard, baseline sequential implementation of the FFT algorithm.
 - Algorithm 2: A parallelized version of the FFT algorithm utilizing C++ threads.
 - We will perform a performance analysis comparing our Algorithm 1 implementation against the Algorithm 2.
- *Idea 2:*
 - Algorithm 1: A parallelized FFT algorithm utilizing threads with a static workload distribution model.
 - Algorithm2 : A parallelized FFT algorithm utilizing threads with dynamic work or task scheduling.
 - We will perform a performance analysis comparing our Algorithm 1 implementation against the Algorithm 2.